

Schema List

1. **Users**(**user_id**, full_name, email, user_type)
2. **Vehicles**(**license_plate**, make_model, owner_id FK -> Users(user_id))
3. **Permits**(**permit_id**, issue_date, expiry_date, permit_type, license_plate FK -> Vehicles(license_plate))
4. **Lots**(**lot_id**, lot_name, zone_type)
5. **Spots**(**spot_id**, spot_number, current_status, lot_id FK -> Lots(lot_id))
6. **Reservations**(**res_id**, start_time, end_time, user_id FK -> Users(user_id), spot_id FK -> Spots(spot_id))
7. **Payments**(**payment_id**, amount, payment_method, transaction_status, processed_at, res_id FK -> Reservations(res_id))
8. **Tickets**(**ticket_id**, violation_reason, fine_amount, is_paid, license_plate FK -> Vehicles(license_plate))

Normalization

Vehicles:

- **Functional Dependency:** license_plate depends on make_model and owner_id.
- **Problem:** If we didn't normalize and kept full_name in the Vehicles table, it could lead to a having to change every instance of that full_name, which is a problem for when the user wants to change their name. So we then use owner_id
- **Solution:** By using owner_id to link to the **Users** table, the name is stored in exactly one place to avoid making the database run operations we can avoid. We use owner_id because it doesn't really change unlike the full_name.

Spots:

- **Functional Dependency:** spot_id -> spot_number, current_status, lot_id.
- **Anomaly:** If we kept lot_name inside the Spots table, we would have a deletion problem. If we deleted all spots in a specific lot, we would lose a way to remember that the lot actually existed.
- **Avoidance:** By moving Lot details to a separate **Lots** table, the Lot data exists independently of the individual spots.

Naming Conventions

Table Names: UPPERCASE with space. EX: LOTS, USERS DATA, ...

Column Names: lowercase, snake_case. EX: owner_id, start_time, ...

Primary Keys: lowercase, snake_case, <appropriate word>_id. EX: license_plate_id for the VEHICLE table.

Foreign Keys: lowercase, snake_case, <appropriate word>_<another appropriate_word>. EX: owner_id, lot_id, ...